

Bounded Almost-Sure Noise for Scalar Thresholding

Research Architecture Runtime

operator/thresholding

2026-07-29

Abstract

Pathwise threshold preservation under almost-sure $|\xi| \leq \eta$ reduces to the deterministic buffer theorem with $\varepsilon := \eta$. Lean `LEAN_FULLL`.

This theorem is: Primitive

1 Problem

The noisy threshold $\tilde{A}_T(x) = \mathbf{1}\{x + \xi \geq T\}$ with $|\xi| \leq \eta$ a.s. The selected object is the binary pass/fail bit.

2 Stability notion

Noise is query-side and bounded. Outside $[T - \eta, T + \eta)$ the output is a.s. constant; inside the band it need not be.

3 Definitions

Pass convention: $x \geq T$ passes. Pathwise: apply deterministic threshold preservation with $\varepsilon = \eta$.

4 Theorem

Theorem 1. *If $|\xi| \leq \eta$ a.s., then $x \geq T + \eta \Rightarrow \tilde{A}_T = 1$ a.s. and $x < T - \eta \Rightarrow \tilde{A}_T = 0$ a.s. On the open band with $\eta > 0$, two-point noise shows the bit need not be a.s. constant.*

5 Intuition

Bounded noise cannot cross a buffer of width η on either side of T .

6 Examples

Example 2. $T = 3$, $x = 5$, $\eta = 0.2$: $x \geq T + \eta$, so the bit stays 1 a.s.

7 Proof

On almost every outcome, $|\xi(\omega)| \leq \eta$. Apply deterministic threshold preservation to $(x, T, \varepsilon, x') = (x, T, \eta, x + \xi(\omega))$: if $x \geq T + \eta$ then $x + \xi \geq T$; if $x < T - \eta$ then $x + \xi < T$. For the band claim, the two-point law on $\{\pm\eta\}$ realizes both sides of the cut (Theorem 1). This is a pathwise surrogate of a.s. bounded noise, not a differential-privacy accounting proof.

8 Formal statement

Lean module

`Research.Operators.Threshold.BoundedNoise`

Source file

`lean/Research/Operators/Threshold/BoundedNoise.lean`

Theorems

- `bounded\_noise\_preservation`
- `bounded\_noise\_sharpness`

Certificate

`lean/certificates/thresholding/bounded-noise-threshold`

Status

`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- `Deterministic.Threshold.Preservation.`
- Pathwise reduction; no DP claim.

10 Consequences

Use for a.s. bounded query noise on scalar thresholds; not Sparse Vector.